Summary Report – Social Media Database (SQL)

This project focuses on designing and analyzing a simple **social media database** using SQL. The database consists of three main tables: **users**, **posts**, and **followers**, representing core functionalities of a social media platform.

The **users table** stores basic user information such as username and email. The **posts table** records user-generated content along with timestamps, while the **followers table** captures relationships between users, enabling a follower-following system.

Sample data was inserted into the tables to simulate real-world scenarios, including different users, posts (with some missing captions), and email domains.

Several SQL queries were performed to analyze the data:

- Filtering users whose names start with a specific letter
- Extracting posts within a given date range
- Identifying posts with missing (NULL) captions
- Categorizing users based on email domains

Key Insights:

- Some users have incomplete data (NULL captions), highlighting real-world data issues
- Users can be grouped based on email providers (Gmail, Yahoo, Hotmail)
- Pattern-based filtering (LIKE) helps in targeted data extraction
- Date-based queries allow tracking of activity over time

Conclusion:

This project demonstrates fundamental SQL skills, including database design, data insertion, and data filtering techniques. It provides a strong foundation for working with real-world applications such as social media platforms and prepares for more advanced database operations.